

Pythia's Advice¹

Floris Padt

Ieke Le Blanc

May 28, 2025

¹Mentor: {} - EAISI Academy, MASTERING DATA & AI, Technical University Eindhoven



Table of Contents



Chapter 1

Preface



Acknowledgments

I am grateful for the insightful comments offered by **Ieke le Blanc**.

These have improved this study in innumerable ways and saved me from many errors; those that inevitably remain are entirely my own responsibility.

1.1 How to Read This Book

This guide will help you navigate through this book and make the most of its features.

1.1.1 Book Structure

This book is organized into the following main sections:

- **Preface:** acknowledgments and how-to read this book



- **Main Research:** exploration of each phase of the CRISP-DM methodology in the [Research Analysis](#)
- **Appendix:** coding, references, and supplementary materials on each CRISP-DM phase.

1.1.2 Navigation Features

This book includes several features to enhance your reading experience:

Table of Contents

The sidebar on the left provides a full table of contents. Click on any section to navigate directly to it.

Search Function

Use the search icon () in the top navigation bar to search for specific terms throughout the book.

Code Blocks

```
x <- 41 # code line 1
x <- 42 # code line 2
```

- Code blocks have a copy button () in the top-right corner
- Some code blocks can be expanded to show more details

Cross-References

This book contains cross-references to:

- figures (Figure ??)
- definitions (def ??)
- sections ([Research Analysis](#))

Interactive Elements

Some sections contain interactive elements:

- Click on images to enlarge them
- Interactive visualizations allow you to explore data
- Collapsible sections can be expanded for additional details

1.1.3 Reading Pathways

1. Start with the [Research Analysis](#) section which is designed to be read in a linear fashion following the CRISP-DM phases.
2. In the Appendix the related coding and detailed analysis can be found on each phase of CRISP-DM, e.g. [Business Understanding](#)



1.1.4 Best Viewing Experience

For the optimal experience:

- Use a modern web browser (Chrome, Firefox, Safari, or Edge)
- Enable JavaScript for interactive features
- Use the dark/light mode toggle for your preferred reading environment

1.1.5 Downloading and Sharing

- Access the source code and data in GitHub with the github icon.
- The pdf icon will open the book as a PDF.
- In section Chapter ?? the pptx and other resources can be locally saved().
- Use the share buttons to share specific chapters on social media.

def 1.1. DEF:

This is an example of a definition $DEF = \sqrt{mean(e_t^2)}$



Part I

Main Research

Chapter 2

Prophecy to Prediction

The Myth Inspiring Pythia's Advice

From Prophecy to Prediction: The Myth Inspiring Pythia's Advice

The *Pythia's Advice* project is dedicated to the art of sales forecasting. Drawing inspiration from the ancient myth of the Oracle of Delphi, where the *Pythia*—the high priestess—delivered cryptic prophecies, we harness the power of Data Science—our modern-day Pythia—to foresee future trends. Just as the Pythia's prophecies required careful interpretation, our statistical and machine learning techniques produce *Advice* that requires human expertise for effective application. By embodying this union of technology and human insight, we aptly named our project *Pythia's Advice*. This synergy ensures well-informed, data-driven decision-making and enhances the operational efficiency of the supply chain.

According to ancient myth, the Pythia inhaled vapors emanating from the remnants of the Python slain by Apollo—the Olympian god of the sun, music, and prophecy. These vapors induced a trance-like state that allowed her to channel his prophetic insights, revealing future events. Similarly, our project 'inhales' vast amounts of data—our 'Python's Vapors'—which, when processed with complex algorithms, enable us to decipher hidden patterns and unveil valuable foresight. However, these predictive insights must be carefully interpreted to avoid missteps—much like when King Croesus misread the Pythia's prophecy that '*a great empire will fall.*' He waged war and, in doing so, opened Pandora's Box—leading to the downfall of his own *great empire*—the Kingdom of Lydia.

Furthermore, *Pythia's Advice* emphasizes the crucial bond between humans and nature. By focusing on the sales forecast of *food for biodiversity*—products that support ecological diversity—we aim to benefit human health and promote planetary well-being. This harmonious relationship mirrors the Pythia's sacred connection with the divine, symbolizing a balance between technological capabilities and the natural world.



Our logo, featuring a tree integrated with the *Pythia* through conductive traces against an enlightening sunset that symbolizes guidance and inspiration, beautifully encapsulates this synergy. This imagery symbolizes the fusion of natural wisdom and technological innovation. It reminds us that while we rely on cutting-edge technology to predict the future, our roots remain deeply intertwined with nature.

Through *Pythia's Advice*, we honor the legacy of the Oracle of Delphi, blending ancient wisdom with modern technology to illuminate the path forward.

Chapter 3

Research Analysis

TRUST - is all you need!



Chapter 4

Introduction

In an increasingly complex market environment, accurate demand forecasting is critical for ensuring the operational efficiency of supply chains, particularly in industries like wholesale, where products are perishable and sales are affected by seasonality and promotions. This research focuses on a wholesale company that buys, produces, and distributes biodiversity food to retailers, Out-Of-Home, discounters and E-commerce. To make informed decisions regarding production, buying, and negotiation with customers, accurate sales forecasts are required at different aggregation levels, depending on each use case.

While the company employs robust but dated statistical methods, such as moving averages and exponential smoothing, these techniques struggle to handle promotional impacts, capture complex sales patterns, and provide prediction intervals. Currently, a combination of bottom-up and top-down forecasting approaches is used, largely determined by the judgment of individual demand planners. However, the lack of clear guidance on which approach to apply in different scenarios has resulted in inconsistent and subjective forecasting outcomes. Furthermore, various use cases have distinct aggregation and time horizon requirements, adding complexity to maintaining consistent and accurate forecasts across the system.

The primary objective of this research is to determine the best way to generate a forecast, which methods to use and on what levels of granularity, enabling consistent, accurate forecasts that can be aggregated or dis-aggregated as required for different business needs. Specifically, the study aims to address the following key questions:

1. Which robust forecasting methods—statistical or data science techniques—should be used to meet the needs of each use case?
2. Which levels of aggregation provides the most accurate forecast for different use cases, and what when is an additional level justified by increased forecast accuracy.
3. What forecast accuracy improvements can be achieved through the integration of additional data, and what are the trade-offs or costs associated with these enhancements?
4. How should human resources be allocated effectively, given the large number of products (SKUs) and varying levels of importance across these products?

Ultimately, this research will provide an objective, data-driven strategy to improve forecast accuracy, reduce reliance on individual intuition, and ensure that forecasts are robust, consistent, and scalable across multiple aggregation levels, thereby enhancing overall decision-making efficiency.



4.1 Methodology

This research follows the **CRISP-ML methodology**, (Costa, 2022) as the guiding framework.

CRISP-ML is an acronym for “Cross-Industry Standard Process for Machine Learning.” It is a systematic framework for organizing and executing machine learning projects. The methodology includes six key steps:

1. understanding the problem
2. preparing the data
3. selecting and tuning models
4. evaluating performance
5. deploying the solution
6. monitoring and maintaining the model.

This thesis focuses on the first 4 steps, the results will be taken as advice to be implemented in the current way of working.

Chapter 5

Business Understanding

where the business process is mapped and when the main indicators are identified, as well as when the business objectives are defined What is the business's need?

Figure 5.1: Why forecasting?
The forecast drives the business!



See Figure Figure ?? for the thumbnail overview.



Chapter 6

Data Understanding



Chapter 7

Data Preparation



Chapter 8

Modeling



Chapter 9

Evaluation

1.2

floris@smart-r.nl

Note

Note that there are five types of callouts, including: **note**, **warning**, **important**, **tip**, and **caution**.

Tip with Title

This is an example of a callout with a title.

Expand To Learn About Collapse

This is an example of a ‘folded’ caution callout that can be expanded by the user. You can use **collapse="true"** to collapse it by default or **collapse="false"** to make a collapsible callout that is expanded by default.



Chapter 10

Deployment



Chapter 11

Conclusion



Glossary



Chapter 12

Conclusions

Conclusions subtitle

3%



Part II

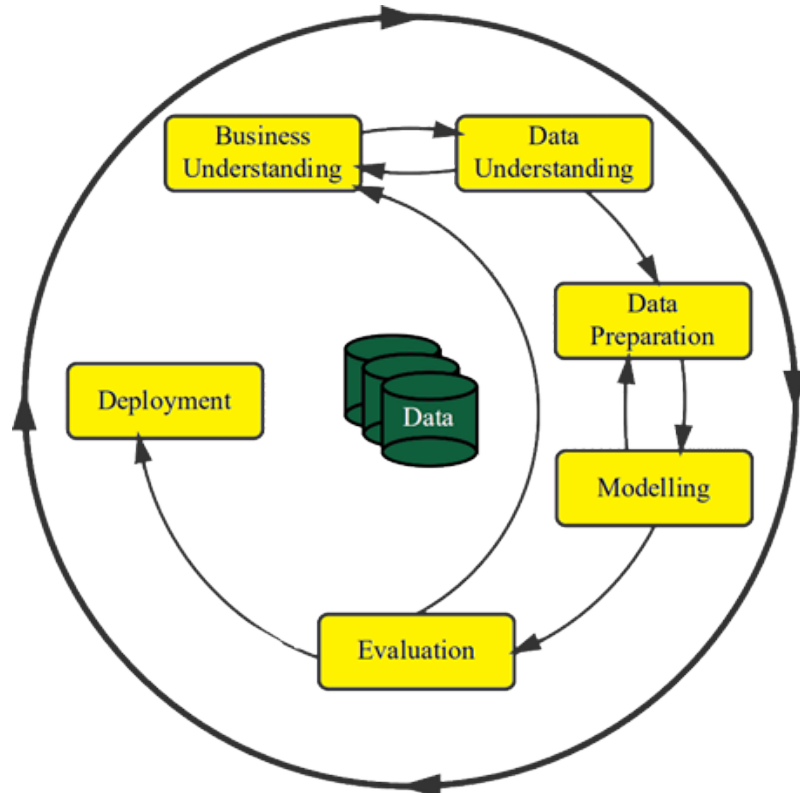
Appendix

Chapter 13

CRISP-DM

13.1 CRISP-DM

Figure 13.1: Project Plan





Chapter 14

Business Understanding

What is the business need?

Figure 14.1: Why forecasting?
The forecast drives the business!





Shipments by Origin → Destination
Metrics grouped by origin using GT row groups

Metric	TO					
	BE	DE	FR	GB	IT	NL
3P						
<i>Count (n)</i>	2	8	13	7	5	11
<i>Average Items</i>	14.0	18.9	97.1	17.9	10.6	18.7
<i>Max Items</i>	19	77	402	44	20	52
DE						
<i>Count (n)</i>	NA	4	3	2	2	3
<i>Average Items</i>	NA	58.5	21.0	4.5	3.5	17.3
<i>Max Items</i>	NA	135	32	7	5	32
ES						
<i>Count (n)</i>	NA	NA	3	NA	2	NA
<i>Average Items</i>	NA	NA	2.3	NA	5.5	NA
<i>Max Items</i>	NA	NA	5	NA	6	NA
FR						
<i>Count (n)</i>	NA	NA	1	1	NA	1
<i>Average Items</i>	NA	NA	152.0	4.0	NA	10.0
<i>Max Items</i>	NA	NA	152	4	NA	10
GB						
<i>Count (n)</i>	NA	1	1	1	2	2
<i>Average Items</i>	NA	61.0	84.0	163.0	23.5	85.5
<i>Max Items</i>	NA	61	84	163	45	168
IT						
<i>Count (n)</i>	1	1	1	1	1	1
<i>Average Items</i>	1.0	22.0	155.0	3.0	194.0	25.0
<i>Max Items</i>	1	22	155	3	194	25



Shipments by Origin → Destination
Manual row grouping with pure data.table syntax

Metric	TO					
	BE	DE	FR	GB	IT	NL
IT						
<i>Average Items</i>	1.0	22.0	155.0	3.0	194.0	25.0
<i>Count (n)</i>	1	1	1	1	1	1
<i>Max Items</i>	1	22	155	3	194	25
GB						
<i>Average Items</i>	NA	61.0	84.0	163.0	23.5	85.5
<i>Count (n)</i>	NA	1	1	1	2	2
<i>Max Items</i>	NA	61	84	163	45	168
FR						
<i>Average Items</i>	NA	NA	152.0	4.0	NA	10.0
<i>Count (n)</i>	NA	NA	1	1	NA	1
<i>Max Items</i>	NA	NA	152	4	NA	10
ES						
<i>Average Items</i>	NA	NA	2.3	NA	5.5	NA
<i>Count (n)</i>	NA	NA	3	NA	2	NA
<i>Max Items</i>	NA	NA	5	NA	6	NA
DE						
<i>Average Items</i>	NA	58.5	21.0	4.5	3.5	17.3
<i>Count (n)</i>	NA	4	3	2	2	3
<i>Max Items</i>	NA	135	32	7	5	32
3P						
<i>Average Items</i>	14.0	18.9	97.1	17.9	10.6	18.7
<i>Count (n)</i>	2	8	13	7	5	11
<i>Max Items</i>	19	77	402	44	20	52



Shipments by Origin → Destination
Origin grouping displayed in left column

FROM	Metric	TO					
		BE	DE	FR	GB	IT	NL
3P	<i>Average Items</i>	14.0	18.9	97.1	17.9	10.6	18.7
	<i>Count (n)</i>	2	8	13	7	5	11
	<i>Max Items</i>	19	77	402	44	20	52
DE	<i>Average Items</i>	NA	58.5	21.0	4.5	3.5	17.3
	<i>Count (n)</i>	NA	4	3	2	2	3
	<i>Max Items</i>	NA	135	32	7	5	32
ES	<i>Average Items</i>	NA	NA	2.3	NA	5.5	NA
	<i>Count (n)</i>	NA	NA	3	NA	2	NA
	<i>Max Items</i>	NA	NA	5	NA	6	NA
FR	<i>Average Items</i>	NA	NA	152.0	4.0	NA	10.0
	<i>Count (n)</i>	NA	NA	1	1	NA	1
	<i>Max Items</i>	NA	NA	152	4	NA	10
GB	<i>Average Items</i>	NA	61.0	84.0	163.0	23.5	85.5
	<i>Count (n)</i>	NA	1	1	1	2	2
	<i>Max Items</i>	NA	61	84	163	45	168
IT	<i>Average Items</i>	1.0	22.0	155.0	3.0	194.0	25.0
	<i>Count (n)</i>	1	1	1	1	1	1
	<i>Max Items</i>	1	22	155	3	194	25

table6, delete?

14.1.1.1 hybrid grouping



Shipments by Origin → Destination
Enhanced left-column grouping with visual indicators

FROM	Metric	TO					
		BE	DE	FR	GB	IT	NL
3P	Average Items	14.0	18.9	97.1	17.9	10.6	18.7
	Count (n)	2	8	13	7	5	11
	Max Items	19	77	402	44	20	52
DE	Average Items	NA	58.5	21.0	4.5	3.5	17.3
	Count (n)	NA	4	3	2	2	3
	Max Items	NA	135	32	7	5	32
ES	Average Items	NA	NA	2.3	NA	5.5	NA
	Count (n)	NA	NA	3	NA	2	NA
	Max Items	NA	NA	5	NA	6	NA
FR	Average Items	NA	NA	152.0	4.0	NA	10.0
	Count (n)	NA	NA	1	1	NA	1
	Max Items	NA	NA	152	4	NA	10
GB	Average Items	NA	61.0	84.0	163.0	23.5	85.5
	Count (n)	NA	1	1	1	2	2
	Max Items	NA	61	84	163	45	168
IT	Average Items	1.0	22.0	155.0	3.0	194.0	25.0
	Count (n)	1	1	1	1	1	1
	Max Items	1	22	155	3	194	25



Chapter 15

GGPlots

ggplot and spark



Lead Time Analysis by Category
Sparkline distributions with performance indicators

Category	Count	Mean	Median	Std Dev	Min	Max	95th %ile	Distribution
Non category specific	4	3.0	3.5	1.4	1.0	4.0	4.0	
Functional Foods	5	7.4	5.0	8.4	1.0	22.0	18.8	
Others	5	10.4	13.0	7.1	1.0	17.0	16.8	
Breakfast Cereals	10	15.1	4.5	17.2	1.0	45.0	42.3	
Bread and Biscuit Alternatives	5	19.4	19.0	13.2	2.0	35.0	33.8	
Sweet Spreads, Purees, Sweeteners	7	24.9	16.0	40.3	2.0	115.0	85.6	
Bread and Bakery	1	30.0	30.0	NA	30.0	30.0	30.0	
Juices and Cold Drinks	3	36.3	34.0	27.6	10.0	65.0	61.9	
Sweets in Between	8	52.8	27.0	87.8	2.0	267.0	188.9	
Dairy Alternatives	10	53.8	21.0	70.2	1.0	194.0	176.4	
Hot Drinks	13	64.8	51.0	59.4	4.0	168.0	165.0	
Meals and Meal components	9	79.1	22.0	128.3	1.0	402.0	295.2	
Animal based products	1	123.0	123.0	NA	123.0	123.0	123.0	

Chapter 16

Sparklines

Update Sparklines

